

Quiz 10.4 – Bond Enthalpies

Name: Key

Problem 1 (4 points)

Give the approximate ΔH_{rxn} for this reaction: $N_2(g) + O_2(g) \rightarrow 2NO(g)$

You may need to use the following values

Average Bond Enthalpies					
Compound	$\Delta H_{Bond} \left(\frac{kJ}{mol} \right)$	Compound	$\Delta H_{Bond} \left(\frac{kJ}{mol} \right)$	Compound	$\Delta H_{Bond} \left(\frac{kJ}{mol} \right)$
O=O	498	O-O	142		
N-N	240	N=N	418	N≡N	941
N-O	200	N=O	607		



Bonds Broken



Bonds Formed



$$\Delta H_{rxn} = 941 \frac{kJ}{mol} + 498 \frac{kJ}{mol} - 2 \times 607 \frac{kJ}{mol} = 225 \frac{kJ}{mol}$$

Problem 2 (1 point)

Briefly explain why this value is only an *approximation* of the true value

Bond strength is affected by the local environment, so not every N=O bond has the same enthalpy. Values in this table are only averages taken from a slate of representative molecules.